

CM0845 Logic

Propositional Logic: Satisfiability

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2016-1

Preliminaries

Textbook

Mordechai Ben-Ari (2012). Mathematical Logic for Computer Science. Section 2.5.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Satisfiability, Validity, Unsatisfiability and Falsifiability

Definition 2.38

Let $\varphi \in \text{PROP}$.

- (i) φ is **satisfiable** iff $\llbracket \varphi \rrbracket_v = 1$ for some interpretation v .
In this case, v is called a model for φ .
- (ii) φ is **valid** (a tautology), denoted $\models \varphi$, iff $\llbracket \varphi \rrbracket_v = 1$ for all interpretations v .
- (iii) φ is **unsatisfiable** iff it is not satisfiable, that is, if $\llbracket \varphi \rrbracket_v = 0$ for all interpretations v .
- (iv) φ is **falsifiable**, denoted $\not\models$, iff it is not valid, that is, if $\llbracket \varphi \rrbracket_v = 0$ for some interpretation v .

Satisfiability, Validity, Unsatisfiability and Falsifiability

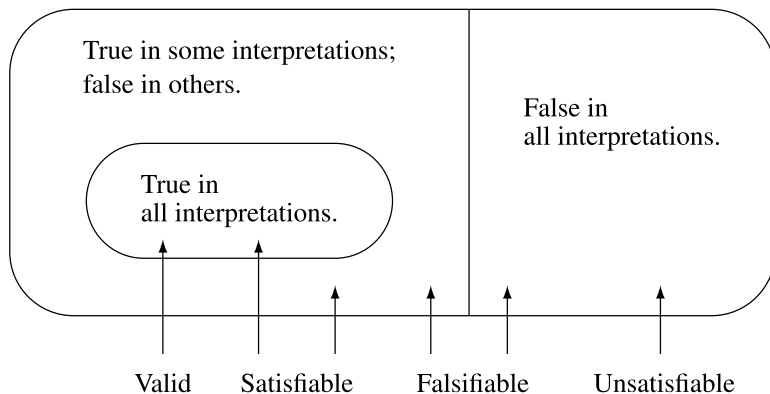


Figure 2.6.

Satisfiability, Validity, Unsatisfiability and Falsifiability

Theorem 2.39

Let $\varphi \in \text{PROP}$.

- (i) The proposition φ is valid if and only if $\neg\varphi$ is unsatisfiable.
- (ii) The proposition φ is satisfiable if and only if $\neg\varphi$ is falsifiable.

Satisfiability of a Set of Propositions

Definition 2.42

Let $\Gamma = \{\varphi_1, \dots\}$ be a set of propositions.

- (i) Γ is **satisfiable** iff there exists an interpretation v such that $\llbracket \varphi \rrbracket_v = 1$ for all $\varphi_i \in \Gamma$.
In this case, v is a model of Γ .
- (ii) Γ is **unsatisfiable** iff for every interpretation v , there exists an $\varphi_i \in \Gamma$ such that $\llbracket \varphi \rrbracket_v = 0$.

Satisfiability of a Set of Propositions

Definition 2.42

Let $\Gamma = \{\varphi_1, \dots\}$ be a set of propositions.

- (i) Γ is **satisfiable** iff there exists an interpretation v such that $\llbracket \varphi \rrbracket_v = 1$ for all $\varphi_i \in \Gamma$.
In this case, v is a model of Γ .
- (ii) Γ is **unsatisfiable** iff for every interpretation v , there exists an $\varphi_i \in \Gamma$ such that $\llbracket \varphi \rrbracket_v = 0$.

Example (Exercise 2.15)

Prove that if Γ is unsatisfiable and for some i , the proposition φ_i is valid, then $\Gamma - \{\varphi_i\}$ is unsatisfiable.

References

Mordechai Ben-Ari [1993] (2012). Mathematical Logic for Computer Science. 3rd ed. Springer (cit. on p. 2).